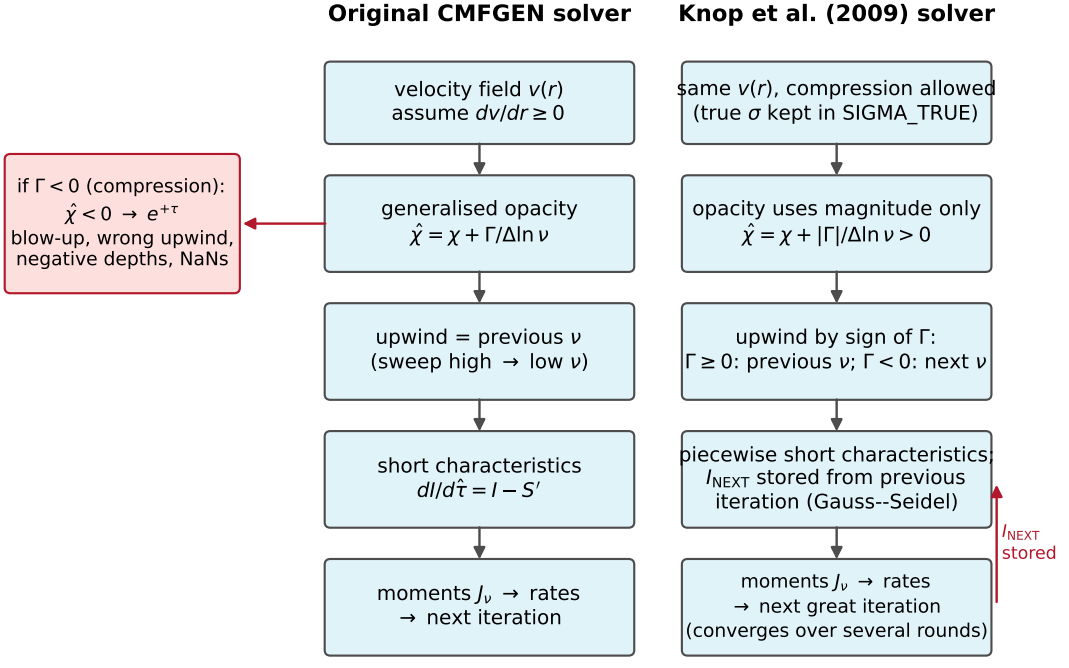

Original CMFGEN solver
Knop et al. (2009) solver
velocity field $v(r)$
assume $dv/dr \geq 0$
same $v(r)$, compression allowed
(true σ kept in SIGMA_TRUE)
if $\Gamma < 0$ (compression):
$\hat{\chi} < 0 \rightarrow e^{+\tau}$
blow-up, wrong upwind,
negative depths, NaNs
generalised opacity
$\hat{\chi} = \chi + \Gamma/\Delta \ln \nu$
opacity uses magnitude only
$\hat{\chi} = \chi + |\Gamma|/\Delta \ln \nu > 0$
upwind = previous ν
(sweep high $\rightarrow$ low ν)
upwind by sign of Γ:
$\Gamma \geq 0$: previous ν; $\Gamma < 0$: next ν
short characteristics
$dI/d\hat{\tau} = I - S'$
piecewise short characteristics;
I_{NEXT} stored from previous
iteration (Gauss--Seidel)
I_{NEXT}
stored
moments $J_\nu \rightarrow$ rates
$\rightarrow$ next iteration
moments $J_\nu \rightarrow$ rates
$\rightarrow$ next great iteration
(converges over several rounds)